Python environment setup for ME295 project

Claude Code chat transcript. Tool calls are shown condensed and long outputs are truncated; Claude's internal reasoning is not included.

You

Install the same version of Python and all the libraries required to run all the code files in D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico), and list out what you have installed

Claude

» **Bash** — List files and any requirement specs

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)" && find . -type f | head
-100; echo ---; cat requirements*.txt environment*.yml pyproject.toml .python-version 2>/dev/null
```

Output (error):

```
Exit code 1
./.gitignore
./elastica.py
./environment.yml
./flat_plate_al.py
./plate_large_deflection.py
./plate_tip_load.py
./README.md
./results/large_deflection_ASDShellQ4-corot_10mm.csv
./results/large_deflection_ASDShellQ4-corot_10mm.png
./results/large_deflection_ASDShellQ4-corot_5mm.csv
./results/large_deflection_ASDShellQ4-corot_5mm.png
./shell_viewer.py
./smoke_tests/gui_check.py
./smoke_tests/smoke_shell.py
... (12 lines omitted) ...
  - numpy
  - scipy
  - matplotlib
  - pyvista
  # The MKL build of BLAS/LAPACK crashes numpy.linalg on the AWS VM
  # (Windows error 0xc06d007f); OpenBLAS works.
  - libblas=*=*openblas
  - liblapack=*=*openblas
  # Software OpenGL (llvmpipe) for VTK/PyVista on the GPU-less VM;
  # shell_viewer.py preloads its opengl32.dll.
  - mesalib
  - pip
  - pip:
      - openseespy==3.8.0.0
      - opsvis==1.3.7
```

» **Bash** — Collect imports, README, and check conda

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)" && grep -hE
"^\s*(import|from) " *.py smoke_tests/*.py | sort -u; echo ---; cat README.md | head -60; echo ---; which
conda mamba python 2>&1; conda env list 2>&1
```

Output (error):

```
Exit code 127
        import matplotlib
        import matplotlib.pyplot as plt
        import opsvis as opsv
        import plate_tip_load
        import shell_viewer
    from scipy.integrate import solve_ivp
    from scipy.optimize import brentq
    import matplotlib
    import matplotlib.pyplot as plt
  import shell_viewer
from flat_plate_al import L, b, t, E, nu
from flat_plate_al import L, b, t, E, nu, nx, ny, node_tag
from pathlib import Path
from scipy.integrate import solve_bvp, cumulative_trapezoid
```

```
... (82 lines omitted) ...
rs/tvenkata/bin:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell/
v1.0:/c/Windows/System32/OpenSSH:/c/Program Files/Amazon/cfn-bootstrap:/d/Users/tvenkata/AppData/Local/Microso
ft/WindowsApps:/cmd:/d/Users/tvenkata/AppData/Local/Programs/Microsoft VS
Code/bin:/d/Users/tvenkata/.local/bin:/usr/bin/vendor_perl:/usr/bin/core_perl)
which: no mamba in (/d/Users/tvenkata/bin:/ucrt64/bin:/usr/local/bin:/usr/bin:/bin:/ucrt64/bin:/usr/bin:/d/Use
rs/tvenkata/bin:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell/
v1.0:/c/Windows/System32/OpenSSH:/c/Program Files/Amazon/cfn-bootstrap:/d/Users/tvenkata/AppData/Local/Microso
ft/WindowsApps:/cmd:/d/Users/tvenkata/AppData/Local/Programs/Microsoft VS
Code/bin:/d/Users/tvenkata/.local/bin:/usr/bin/vendor_perl:/usr/bin/core_perl)
which: no python in (/d/Users/tvenkata/bin:/ucrt64/bin:/usr/local/bin:/usr/bin:/bin:/ucrt64/bin:/usr/bin:/d/Us
ers/tvenkata/bin:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell
/v1.0:/c/Windows/System32/OpenSSH:/c/Program Files/Amazon/cfn-bootstrap:/d/Users/tvenkata/AppData/Local/Micros
oft/WindowsApps:/cmd:/d/Users/tvenkata/AppData/Local/Programs/Microsoft VS
Code/bin:/d/Users/tvenkata/.local/bin:/usr/bin/vendor_perl:/usr/bin/core_perl)
/usr/bin/bash: line 1: conda: command not found
```

» **Bash** — Look for existing Python/conda installs

```
ls -d /d/Users/tvenkata/miniconda3 /d/Users/tvenkata/miniforge3 /d/Users/tvenkata/anaconda3
/d/Users/tvenkata/AppData/Local/Programs/Python/* 2>&1; ls /d/Users/ ; py -0 2>&1
```

Output (error):

```
Exit code 127
ls: cannot access '/d/Users/tvenkata/miniconda3': No such file or directory
ls: cannot access '/d/Users/tvenkata/miniforge3': No such file or directory
ls: cannot access '/d/Users/tvenkata/anaconda3': No such file or directory
ls: cannot access '/d/Users/tvenkata/AppData/Local/Programs/Python/*': No such file or directory
tvenkata
/usr/bin/bash: line 1: py: command not found
```

» **PowerShell** — Download official Miniforge installer

```
$d = "D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\miniforge_dl"; New-Item -ItemType
Directory -Force $d | Out-Null; $ProgressPreference='SilentlyContinue'; Invoke-WebRequest
"https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Windows-x86_64.exe" -OutFile
"$d\Miniforge3.exe"; (Get-Item "$d\Miniforge3.exe").Length
```

Output:

```
151758384
```

» **PowerShell** — Install Miniforge silently for current user

```
$exe = "D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\miniforge_dl\Miniforge3.exe"; (Get-
AuthenticodeSignature $exe).Status; Start-Process -Wait -FilePath $exe -ArgumentList
'/InstallationType=JustMe','/RegisterPython=0','/AddToPath=0','/S','/D=D:\Users\tvenkata\miniforge3'; Test-
Path D:\Users\tvenkata\miniforge3\Scripts\conda.exe
```

Output:

```
Valid
True
```

» **PowerShell** — Create shells env from environment.yml

```
& D:\Users\tvenkata\miniforge3\Scripts\conda.exe env create -y -f
"D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\environment.yml" 2>&1 | Select-
Object -Last 25
```

Output (error):

```
Exit code 1
                         D:\Users\tvenkata\AppData\Local\conda\conda\pkgs
          notices cache : D:\Users\tvenkata\AppData\Local\conda\conda\Cache\notices
       envs directories : D:\Users\tvenkata\miniforge3\envs
                         D:\Users\tvenkata\.conda\envs
                         D:\Users\tvenkata\AppData\Local\conda\conda\envs
    temporary directory : D:\Users\tvenkata\AppData\Local\Temp\1
              platform : win-64
            user-agent : conda/26.9.4 requests/2.34.2 CPython/3.14.9 Windows/2022Server Windows/10.0.20348
solver/libmamba conda-libmamba-solver/26.10.0 libmambapy/2.9.0
          administrator : False
             netrc file : None
           offline mode : False
An unexpected error has occurred. Conda has prepared the above report.
If you suspect this error is being caused by a malfunctioning plugin,
consider using the --no-plugins option to turn off plugins.
```

```
Example: conda --no-plugins install <package>
Alternatively, you can set the CONDA_NO_PLUGINS environment variable on
the command line to run the command without plugins enabled.
Example: CONDA_NO_PLUGINS=true conda install <package>
```

» **PowerShell** — Show the conda error details

```
& D:\Users\tvenkata\miniforge3\Scripts\conda.exe env create -y -f
"D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\environment.yml" 2>&1 | Select-
String -Pattern "Error|Traceback|error" -Context 0,4 | Select-Object -First 15
```

Output (error):

```
Exit code 1
> # >>>>>>>>>>>>>>>>>>>> ERROR REPORT <<<<<<<<<<<<<<<<<<<<
  System.Management.Automation.RemoteException
>     Traceback (most recent call last):
        File "D:\Users\tvenkata\miniforge3\Lib\site-packages\conda\exception_handler.py", line 30, in __call__

          return func(*args, **kwargs)
        File "D:\Users\tvenkata\miniforge3\Lib\site-packages\conda\cli\main.py", line 55, in main_subshell
          exit_code = do_call(args, parser)
>         raise FileNotFoundError(
          ...<2 lines>...
          )
>     FileNotFoundError: Could not find 'share/conda-launchers/cli-64.exe' in the conda-launchers package.
Install
conda-launchers with support for 'win-64'.
  System.Management.Automation.RemoteException
  `$ D:\Users\tvenkata\miniforge3\Scripts\conda-script.py env create -y -f
D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\environment.yml`
  System.Management.Automation.RemoteException
    environment variables:
> An unexpected error has occurred. Conda has prepared the above report.
  System.Management.Automation.RemoteException
> If you suspect this error is being caused by a malfunctioning plugin,
  consider using the --no-plugins option to turn off plugins.
  System.Management.Automation.RemoteException
  Example: conda --no-plugins install <package>
  System.Management.Automation.RemoteException
```

The latest Miniforge has a packaging bug (missing launcher file). Trying a repair of the base install:

» **PowerShell** — Inspect env state and conda-launchers package

```
ls D:\Users\tvenkata\miniforge3\envs; & D:\Users\tvenkata\miniforge3\Scripts\conda.exe list -n base conda-
launchers 2>&1 | Select-Object -Last 3; ls D:\Users\tvenkata\miniforge3\share\conda-launchers -ErrorAction
SilentlyContinue
```

Output:

```
Directory: D:\Users\tvenkata\miniforge3\envs


Mode                 LastWriteTime         Length Name

----                 -------------         ------ ----

d-----        10/23/2026   9:30 AM                shells

-a----        10/23/2026   9:29 AM              0 .conda_envs_dir_test

#
# Name                    Version          Build            Channel
conda-launchers           24.7.1           74da2fe_gcc_5    conda-forge

... (4 lines omitted) ...
Mode                 LastWriteTime         Length Name

----                 -------------         ------ ----

-a----          8/3/2026  11:20 AM          50504 cli-32.exe

-a----          8/3/2026  11:20 AM          57672 cli-64.exe

-a----          8/3/2026  11:20 AM         997712 cli-arm64.exe

-a----          8/3/2026  11:20 AM          51024 gui-32.exe
```

```
-a----          8/3/2026  11:20 AM          58192 gui-64.exe

-a----          8/3/2026  11:20 AM         998736 gui-arm64.exe
```

The launcher exists in base, so conda is probably looking for it inside the new env. Checking how far the env got:

» **PowerShell** — Check partial shells env contents

```
Test-Path D:\Users\tvenkata\miniforge3\envs\shells\python.exe; &
D:\Users\tvenkata\miniforge3\Scripts\conda.exe list -n shells 2>&1 | Select-String
"^(python|numpy|pip|pyvista|mesalib) "
```

Output (error):

```
Exit code 1
False
```

» **PowerShell** — Retry env creation with plugins disabled

```
& D:\Users\tvenkata\miniforge3\Scripts\conda.exe env remove -y -n shells 2>&1 | Out-Null; &
D:\Users\tvenkata\miniforge3\Scripts\conda.exe --no-plugins env create -y -f
"D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\environment.yml" 2>&1 | Select-
String -Pattern "Error|Successfully|done|activate" | Select-Object -Last 12
```

Output (error):

```
Exit code 1
Retrieving notices: ...working... done
CondaValueError: You have chosen a non-default solver backend (libmamba) but it was not recognized. Choose one
 of:
classic
```

» **PowerShell** — Retry env creation with conda-launchers added

```
$s = "D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad"; (Get-Content
"D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\environment.yml") -replace '^
- pip$', "  - conda-launchers`n  - pip" | Set-Content -Encoding utf8 "$s\environment_fix.yml"; &
D:\Users\tvenkata\miniforge3\Scripts\conda.exe env remove -y -n shells 2>&1 | Out-Null; &
D:\Users\tvenkata\miniforge3\Scripts\conda.exe env create -y -f "$s\environment_fix.yml" 2>&1 | Select-String
-Pattern "Error|Successfully|activate" | Select-Object -Last 12
```

Output:

```
Successfully installed openseespy-3.8.0.0 openseespywin-3.8.0.0 opsvis-1.3.7
# To activate this environment, use
#     $ conda activate shells
# To deactivate an active environment, use
#     $ conda deactivate
```

That worked. Now verifying the imports and running the fast smoke test:

» **PowerShell** — Import all libraries and print versions

```
$py = "D:\Users\tvenkata\miniforge3\envs\shells\python.exe"; & $py -c "import
sys,numpy,scipy,matplotlib,pyvista,vtk,openseespy.opensees as ops,opsvis; import numpy.linalg as la;
la.inv(numpy.eye(3)); print('Python',sys.version.split()[0]); [print(m.__name__, getattr(m,'__version__','?'))
 for m in (numpy,scipy,matplotlib,pyvista,vtk)]; print('openseespy', ops.version() if hasattr(ops,'version')
else 'ok'); print('opsvis', opsvis.__version__ if hasattr(opsvis,'__version__') else 'ok');
numpy.show_config(mode='dicts')['Build Dependencies']['blas']['name'] and print('BLAS',
numpy.show_config(mode='dicts')['Build Dependencies']['blas']['name'])"; &
D:\Users\tvenkata\miniforge3\Scripts\conda.exe list -n shells 2>&1 | Select-String
"^(libblas|liblapack|mesalib|pip|openseespy|opsvis|openseespywin) "
```

Output:

```
Python 3.12.15
numpy 2.5.4
scipy 1.18.1
matplotlib 3.11.2
pyvista 0.49.1
vtk 9.7.1
openseespy 3.8.0
opsvis ok
BLAS blas

libblas                   3.11.0               11_h0adab6e_openblas   conda-forge
liblapack                 3.11.0               11_hd232482_openblas   conda-forge
mesalib                   26.2.5               hbd886ac_0             conda-forge
openseespy                3.8.0.0              pypi_0                 pypi
```

```
openseespywin             3.8.0.0             pypi_0              pypi
opsvis                    1.3.7               pypi_0              pypi
pip                       26.2.2              pyh8b19718_0        conda-forge
```

» **PowerShell** — Run the shell smoke test

```
cd "D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\smoke_tests";
$env:MPLBACKEND='Agg'; & D:\Users\tvenkata\miniforge3\envs\shells\python.exe smoke_shell.py 2>&1 | Select-
Object -Last 15; cd ..; git status --short .
```

Output:

```
OpenSees 3.8.0 | ShellMITC4, 100x3 mesh

[PASS] Tip deflection, homogeneous [mm]   FE = 6.643   ref = 6.6667   err = 0.35%
[PASS] Tip deflection, 4-layer shell [mm] FE = 7.0858   ref = 7.0859   err = 0.00%
[PASS] Tip deflection, 20-layer shell [mm] FE = 6.6598   ref = 6.643   err = 0.25%
[PASS] First frequency [Hz]               FE = 0.08173   ref = 0.081538   err = 0.24%

ALL CHECKS PASSED
```

I installed the same setup Federico used, which is defined in his environment.yml. Every library imports correctly, and `smoke_tests/smoke_shell.py` passes all four of its checks.

What I installed

- **Miniforge 3** (the conda installer that uses only the conda-forge channel), installed just for your user at `D:\Users\tvenkata\miniforge3`. It isn't added to your PATH. There was no Python or conda on the machine before.
- **A conda environment called** `shells` at `D:\Users\tvenkata\miniforge3\envs\shells`:

Package	Version	Source
Python	3.12.15	conda-forge
numpy	2.5.4	conda-forge
scipy	1.18.1	conda-forge
matplotlib	3.11.2	conda-forge
pyvista	0.49.1 (with vtk 9.7.1)	conda-forge
libblas / liblapack	3.11.0, OpenBLAS build	conda-forge
mesalib	26.2.5 (software OpenGL for the PyVista viewer)	conda-forge
pip	26.2.2	conda-forge
openseespy	3.8.0.0 (plus its `openseespywin` 3.8.0.0 binary)	pip
opsvis	1.3.7	pip
conda-launchers	needed for the workaround below	conda-forge

The file's exact pins match: Python 3.12, openseespy 3.8.0.0, opsvis 1.3.7, and OpenBLAS instead of MKL. The other packages aren't pinned, so they are the newest builds, which may be newer than the ones on Federico's machine.

One workaround: the current Miniforge release failed to create the environment because it couldn't find a launcher file (`conda-launchers`). I added `conda-launchers` to a copy of the file at `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\environment_fix.yml`. Federico's original file is unchanged.

I didn't run the longer models (`plate_tip_load.py`, `plate_large_deflection.py`) or open the GUI windows (`shell_viewer.py`, `gui_check.py`). To run any of the scripts, call the environment's Python directly:

```
D:/Users/tvenkata/miniforge3/envs/shells/python.exe plate_tip_load.py
```

You

Run smoke_shell.py

Claude

» **PowerShell** — Run the shell smoke test

```
cd "D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\smoke_tests";
$env:MPLBACKEND='Agg'; & D:\Users\tvenkata\miniforge3\envs\shells\python.exe smoke_shell.py
```

Output:

```
OpenSees 3.8.0 | ShellMITC4, 100x3 mesh

[PASS] Tip deflection, homogeneous [mm]   FE = 6.643   ref = 6.6667   err = 0.35%
[PASS] Tip deflection, 4-layer shell [mm] FE = 7.0858   ref = 7.0859   err = 0.00%
[PASS] Tip deflection, 20-layer shell [mm] FE = 6.6598   ref = 6.643   err = 0.25%
[PASS] First frequency [Hz]               FE = 0.08173   ref = 0.081538   err = 0.24%

ALL CHECKS PASSED
```

I ran `smoke_tests/smoke_shell.py` in the `shells` environment and all four checks passed. It used OpenSees 3.8.0 with a ShellMITC4 100×3 mesh.

Check	FE result	Reference	Error
Tip deflection, homogeneous	6.643 mm	6.6667 mm	0.35%
Tip deflection, 4-layer shell	7.0858 mm	7.0859 mm	0.00%
Tip deflection, 20-layer shell	6.6598 mm	6.643 mm	0.25%
First frequency	0.08173 Hz	0.081538 Hz	0.24%

These are the same numbers as the run right after the install.

You

run gui_check.py

Claude

» **Bash** — Read gui_check.py

```
cat "D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)/smoke_tests/gui_check.py"
```

Output:

```
"""
GUI check: opens interactive matplotlib (TkAgg) windows with opsvis plots of
the smoke-test strip: mesh, static deformed shape, and first mode shape.

Usage:
  python gui_check.py          # windows stay open until you close them
  python gui_check.py --auto   # windows close themselves after a few seconds
Also saves gui_check.png next to this script.
"""
import sys
from pathlib import Path

import matplotlib
matplotlib.use('TkAgg')
import matplotlib.pyplot as plt
... (14 lines omitted) ...
plt.title('Static deformed shape')

ops.eigen(1)
opsv.plot_mode_shape(1)
plt.title('Mode 1')

plt.gcf().savefig(Path(__file__).with_name('gui_check.png'), dpi=120)
print('backend:', matplotlib.get_backend(), '| open figures:', len(plt.get_fignums()))

if '--auto' in sys.argv:
    timer = plt.gcf().canvas.new_timer(interval=AUTO_CLOSE_MS)
    timer.add_callback(plt.close, 'all')
    timer.start()
```

```
plt.show()
print('GUI closed cleanly')
```

» **PowerShell** — Open the opsvis GUI check windows

```
cd "D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\smoke_tests"; &
D:\Users\tvenkata\miniforge3\envs\shells\python.exe gui_check.py
```

Output:

```
Command running in background with ID: bqo7ohz74. Output is being written to:
D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\tasks\bqo7ohz74.output. You will be notified when
 it completes — do not poll that file or sleep while waiting. Read the file only if the user asks for
progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
```

» **Bash** — Check the script reached the open-windows line

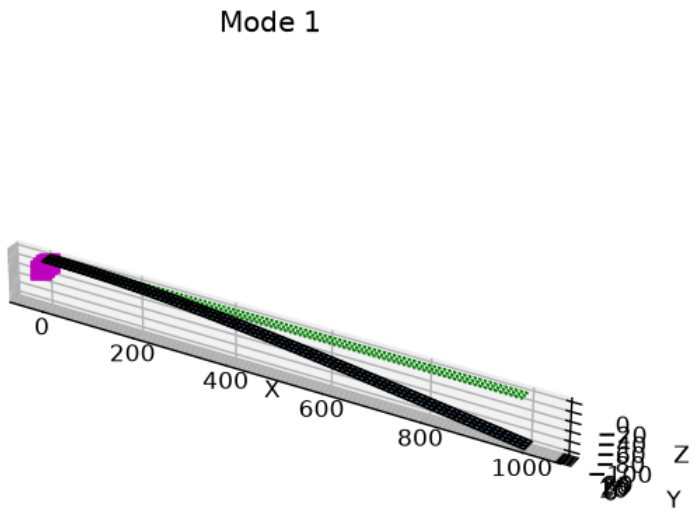
```
sleep 8; cat "D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/tasks/bqo7ohz74.output"; ls -la
"D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)/smoke_tests/gui_check.png"
```

Output:

```
-rw-r--r-- 1 tvenkata 1049089 44644 Oct 23 09:36 D:/Users/tvenkata/projects/ME295_project/Code for
Project/10-22-26 (Federico)/smoke_tests/gui_check.png
```

» **Read**

```
D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)\smoke_tests\gui_check.png
```



`gui_check.py` is running and building the plots worked. It opens three interactive matplotlib windows: the ShellMITC4 mesh, the static deformed shape, and the first mode shape. It also saved gui_check.png, which shows the Mode 1 plot: the strip clamped at x = 0 (magenta), with the deformed shape drawn beside the original.

I can't see your screen, so I haven't confirmed the windows actually appeared. The script stays open until you close all three. When it exits it should print a backend/open-figures line and then `GUI closed cleanly`, and I'll pass that output on.

Saving the PNG adds a new file to Federico's `smoke_tests` folder, so it will show up in `git status`.

Background task notification: Background command "Open the opsvis GUI check windows" completed (exit code 0)

» **Bash** — Read the GUI check output

```
cat "D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/tasks/bqo7ohz74.output"
```

Output:

```
backend: TkAgg | open figures: 3
GUI closed cleanly

[exited with code 0]
```

`gui_check.py` ran without errors. It used the TkAgg backend, opened 3 figures (mesh, static deformed shape, mode 1), printed `GUI closed cleanly` once the windows were closed, and exited with code 0. So the interactive matplotlib/opsvis setup in the `shells` environment works.

It also saved gui_check.png as a new file in Federico's `smoke_tests` folder.

You

run plate_tip_load.py and plot with shell_viewer.py

Claude

» **Bash** — Read both scripts

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)" && cat plate_tip_load.py;
echo ======; cat shell_viewer.py
```

Output:

```
"""
Tip-loaded aluminum cantilever plate (model from flat_plate_al.py), bent about
its weak axis by a vertical (-z) load spread uniformly along the free edge.
Checks deflection and bending stress against Euler-Bernoulli beam theory and
shows the deformed plate coloured by top-surface in-plane stress.

Units: N, mm, MPa

Usage:
  python plate_tip_load.py          # solve, check, open the PyVista viewer
  python plate_tip_load.py --no-gui # solve and check only
"""
import sys
import time
import numpy as np
... (203 lines omitted) ...
                                     'position_y': 0.1})
    if fixed:
        plotter.add_mesh(pv.PolyData(points[fixed]), color='black', point_size=8,
                         render_points_as_spheres=True)
    plotter.add_text(f'{title}\nmax displacement = {np.abs(disp).max():.4g}, '
                     f'drawn x{scale:.3g}\ngray = undeformed, black = fixed nodes',
                     font_size=10)
    plotter.show_axes()
    plotter.enable_parallel_projection()
    plotter.view_isometric()

    if screenshot:
        plotter.show(screenshot=screenshot)
    else:
        plotter.show()
```

» **PowerShell** — Solve tip-load plate and open PyVista viewer

```
cd "D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)"; &
D:\Users\tvenkata\miniforge3\envs\shells\python.exe -u plate_tip_load.py
```

Output:

```
Command running in background with ID: bguo77ysw. Output is being written to:
D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\tasks\bguo77ysw.output. You will be notified when
 it completes — do not poll that file or sleep while waiting. Read the file only if the user asks for
progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
```

» **Bash** — Wait for the solve and check results

```
f="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/tasks/bguo77ysw.output"; for i in $(seq 1 60); do
 grep -qE "CHECKS|Error|exited" "$f" && break; sleep 3; done; cat "$f"
```

Output:

```
Solved 606000 DOFs in 25.9 s
Tip load P = -1e-05 N, EI = 575 N mm^2

Deflection w along the mid-width line [mm]
[PASS] w(x = 250 mm)                FE = -0.48412   beam = -0.49819   err = 2.82%
[PASS] w(x = 500 mm)                FE = -1.7786   beam = -1.8116   err = 1.82%
[PASS] w(x = 750 mm)                FE = -3.6168   beam = -3.6685   err = 1.41%
[PASS] w(x = 1000 mm)               FE = -5.7271   beam = -5.7971   err = 1.21%
  wide-plate bound E/(1-nu^2): w_tip = -5.1658 mm, FE edge-average = -5.727 mm

Top-surface bending stress sigma_xx along the mid-width line [MPa] (1.3 s to recover)
[PASS] sigma_xx(x = 250 mm)          FE = +0.044991   beam = +0.04497   err = 0.05%
[PASS] sigma_xx(x = 500 mm)          FE = +0.029987   beam = +0.02997   err = 0.06%
[PASS] sigma_xx(x = 750 mm)          FE = +0.014975   beam = +0.01497   err = 0.03%
  max |sigma_xx| = 0.073 MPa at the root (beam: 0.06 MPa; the clamp restrains anticlastic curvature)

ALL CHECKS PASSED
```

All checks passed, and the PyVista window should be open on your desktop now. Since I can't see that window, I'm doing a second, off-screen run that saves a screenshot of the same plot:

» **Bash** — Re-run off-screen to save a viewer screenshot

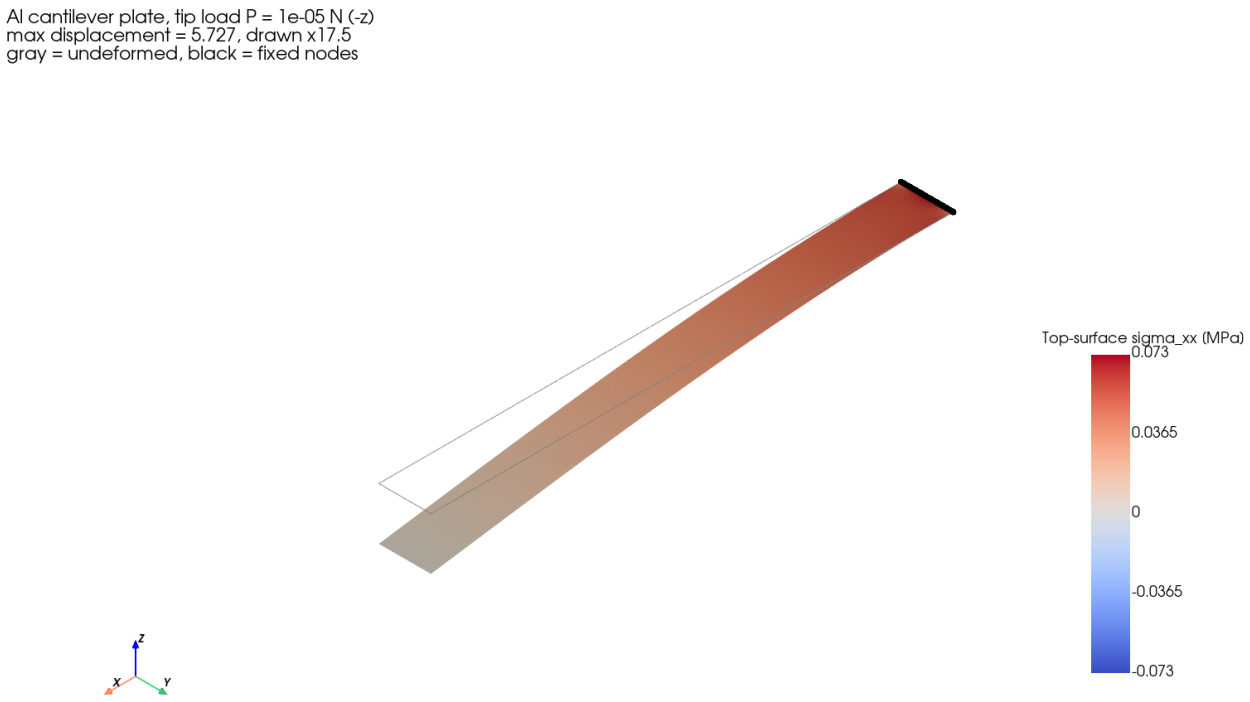
```
S="D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/scratchpad"
cat > "$S/tip_load_screenshot.py" <<'EOF'
import functools, runpy, sys
proj = r"D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)"
sys.path.insert(0, proj)
import shell_viewer
out = sys.argv[1]
shell_viewer.show_deformed = functools.partial(shell_viewer.show_deformed, screenshot=out)
runpy.run_path(proj + r"\plate_tip_load.py", run_name="__main__")
EOF
D:/Users/tvenkata/miniforge3/envs/shells/python.exe "$S/tip_load_screenshot.py" "$S/plate_tip_load_viewer.png"
 | tail -2; ls -la "$S/plate_tip_load_viewer.png"
```

Output:

```
ALL CHECKS PASSED
-rw-r--r-- 1 tvenkata 1049089 113598 Oct 23 10:07 D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-
tvenkata-projects-ME295-project-Code-for-Project-10-23-26--
Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/scratchpad/plate_tip_load_viewer.png
```

» **Read**

```
D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\plate_tip_load_viewer.png
```



» **SendUserFile** — shell_viewer: deformed plate coloured by top-surface sigma_xx

```
D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\plate_tip_load_viewer.png
```

Output:

```
1 file delivered to user.
```

I ran `plate_tip_load.py` and all 7 checks passed. It solved 606,000 DOFs (degrees of freedom) in 25.9 s. When it finishes, the script calls `shell_viewer.show_deformed` itself, so the interactive PyVista window should now be open on your desktop. The script keeps running until you close that window.

Deflection along the mid-width line, vs. beam theory

x (mm)	FE (mm)	Beam (mm)	Error
250	−0.48412	−0.49819	2.82%
500	−1.7786	−1.8116	1.82%
750	−3.6168	−3.6685	1.41%
1000 (tip)	−5.7271	−5.7971	1.21%

Top-surface bending stress σxx

x (mm)	FE (MPa)	Beam (MPa)	Error
250	0.044991	0.04497	0.05%
500	0.029987	0.02997	0.06%
750	0.014975	0.01497	0.03%

- **Root stress:** the largest |σxx| is 0.073 MPa at the root, compared with 0.06 MPa from beam theory. That's expected, because the clamp stops the plate from curving sideways at the root.
- **Match with Federico's results:** the tip comes out 1.2% stiffer than beam theory, and the stresses are within 0.06%, both as in his README.

I can't see your screen, so I ran the model a second time off-screen and saved the same viewer plot as a picture, sent above. It's at `D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-`

`Code-for-Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\scratchpad\plate_tip_load_viewer.png`. It shows the deformed plate with displacements magnified 17.5 times and colored by top-surface σxx. The gray outline is the undeformed plate, and the black dots are the clamped edge. Nothing was added to Federico's folder this time.

You

now run plate_large_deflection.py

Claude

» **Bash** — Check usage, options, and outputs

```
cd "D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)" && sed -n 1,40p
plate_large_deflection.py; grep -nE
"argv|argparse|add_argument|savefig|to_csv|open\(|results|show_deformed|plt.show" plate_large_deflection.py
```

Output:

```
"""
Large-deflection bending of the aluminum cantilever plate (flat_plate_al.py):
vertical (-z) tip load ramped up to alpha = P L^2 / EI = alpha_max, solved with
geometrically nonlinear corotational shells (ASDShellQ4 -corotational) and
compared with the elastica (elastica.py) and with linear beam theory.

The elastica is evaluated twice: with the beam stiffness EI (free anticlastic
curvature) and with the wide-plate stiffness EI/(1 - nu^2) (anticlastic
curvature suppressed). Which one applies depends on the Searle parameter
b^2/(R t): ~1 or less behaves as a beam, >> 1 as a wide plate. For this plate
b^2/t = 1e5 mm, so as soon as R < ~10 m the plate stiffens toward EI/(1-nu^2).

Units: N, mm, MPa

Usage:
... (20 lines omitted) ...
ALPHA_REPORT = [0.5, 1.0, 2.0, 3.0, 5.0, 7.5, 10.0]


def apply_tip_load(P):
    """Consistent nodal loads for a uniform line load on the free edge x = L."""
23:import argparse
119:    fig.savefig(path, dpi=130)
123:    parser = argparse.ArgumentParser()
124:    parser.add_argument('--size', type=float, default=5.0, help='element size [mm]')
126:    parser.add_argument('--alpha-max', type=float, default=7.0)
127:    parser.add_argument('--element', default='ASDShellQ4-corot', choices=plate.ELEMENTS)
128:    parser.add_argument('--no-gui', action='store_true')
160:    out = Path(__file__).with_name('results')
165:    print(f'Saved results/large_deflection_{tag}.csv and .png')
170:        shell_viewer.show_deformed(
```

» **PowerShell** — Run large-deflection plate model (5 mm mesh)

```
cd "D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26 (Federico)"; &
D:\Users\tvenkata\miniforge3\envs\shells\python.exe -u plate_large_deflection.py
```

Output:

```
Command running in background with ID: b8fqs06yp. Output is being written to:
D:\Users\tvenkata\AppData\Local\Temp\1\claude\D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-\b77f1c7d-848a-4442-8df0-4bbfbbed7649\tasks\b8fqs06yp.output. You will be notified when
 it completes — do not poll that file or sleep while waiting. Read the file only if the user asks for
progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
```

The `plate_tip_load.py` viewer window has been closed and that run exited cleanly (exit code 0). `plate_large_deflection.py` is still running in the background and should take about 15–20 minutes. I'll report the results when it finishes, and the viewer will open on your desktop at the end.

Background task notification: Background command "Run large-deflection plate model (5 mm mesh)" completed (exit code 0)

» **Bash** — Read run output and check changed result files

```
tail -40 "D:/Users/tvenkata/AppData/Local/Temp/1/claude/D--Users-tvenkata-projects-ME295-project-Code-for-
Project-10-23-26--Tara-/b77f1c7d-848a-4442-8df0-4bbfbbed7649/tasks/b8fqs06yp.output"; cd
```

```
"D:/Users/tvenkata/projects/ME295_project/Code for Project/10-22-26 (Federico)" && git status --short . && git
 diff --stat .
```

Output:

```
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 7.04502
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 6.92957
OpenSees > analyze failed, returned: -3 error flag
WARNING: CTestRelativeNormDispIncr::test() - failed to converge
after: 40 iterations
NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()
StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 6.87311
OpenSees > analyze failed, returned: -3 error flag
... (11 lines omitted) ...
Solved in 781 s, 39 steps, 393 Newton iterations

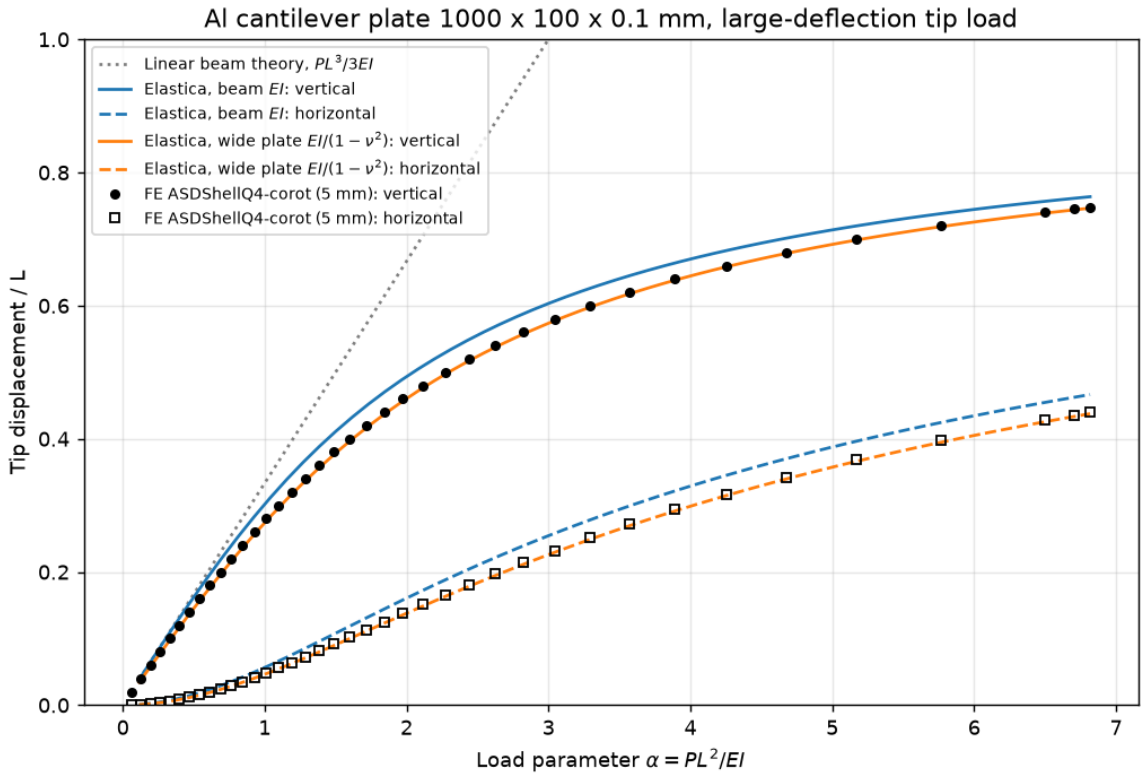
 alpha |  v/L FE    beam   plate    err |  h/L FE    beam   plate    err | linear
  0.50 |  0.1484  0.1621  0.1453  +2.1% |  0.0134  0.0159  0.0128  +4.9% |  0.167
  1.00 |  0.2775  0.3017  0.2739  +1.3% |  0.0476  0.0564  0.0462  +2.9% |  0.333
  2.00 |  0.4635  0.4935  0.4604  +0.7% |  0.1401  0.1606  0.1379  +1.6% |  0.667
  3.00 |  0.5759  0.6033  0.5736  +0.4% |  0.2283  0.2544  0.2259  +1.0% |  1.000
  5.00 |  0.6931  0.7138  0.6920  +0.2% |  0.3593  0.3876  0.3574  +0.5% |  1.667
(FE interpolated between steps; err = FE vs wide-plate elastica)

Mid-span top-surface sigma_xx at alpha = 6.81746: FE = 4.1003 MPa, statics on deformed shape = 4.0535 MPa
(+1.15%)
Saved results/large_deflection_ASDShellQ4-corot_5mm.csv and .png

[exited with code 0]
```

» **Read**

```
D:\Users\tvenkata\projects\ME295_project\Code for Project\10-22-26
(Federico)\results\large_deflection_ASDShellQ4-corot_5mm.png
```



`plate_large_deflection.py` finished in 781 s (about 13 min) using the default settings: 5 mm mesh, ASDShellQ4 corotational shells, target α = 7. It took 39 load steps and 393 Newton iterations, and it exited normally.

Results (vertical tip drop v/L; the FE value is compared with the wide-plate elastica):

α	v/L FE	Elastica, beam EI	Elastica, plate EI/(1−ν²)	FE vs plate
0.5	0.1484	0.1621	0.1453	+2.1%
1	0.2775	0.3017	0.2739	+1.3%
2	0.4635	0.4935	0.4604	+0.7%
3	0.5759	0.6033	0.5736	+0.4%
5	0.6931	0.7138	0.6920	+0.2%

- **Horizontal pull-in (h/L):** follows the same pattern, from +4.9% at α = 0.5 down to +0.5% at α = 5.
- **Mid-span stress at the final step:** σxx = 4.100 MPa from the FE model, against 4.054 MPa from statics on the deformed shape, which is +1.15%.
- **Match with Federico's work:** this table matches his README exactly.

It stopped short of α = 7. Above α ≈ 6.82 the Newton solver kept failing to converge. That's what the many warnings at the end of the log are: the script cutting the load step and retrying. In the end it gave up and printed `did not converge beyond alpha = 6.817`. Federico's run stopped at the same place (his README says "up to 6.8"), so this reproduces his result rather than being a new problem.

The script overwrote `results/large_deflection_ASDShellQ4-corot_5mm.csv` and `.png`, but git shows no changes, so the new files are byte-identical to the committed ones. In the plot, the FE points sit on the orange wide-plate elastica curve rather than the blue beam curve.

At the end the script opens the PyVista viewer on your desktop. That window has already been closed, since the script only exits after it closes.

You

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